

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME:Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)

Total Marks: 20

Question Count: 4

Code of Conduct

I Ch.Tharunika Reddy(192324307) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ch.Tharunika Reddy

Assessment Tasks Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

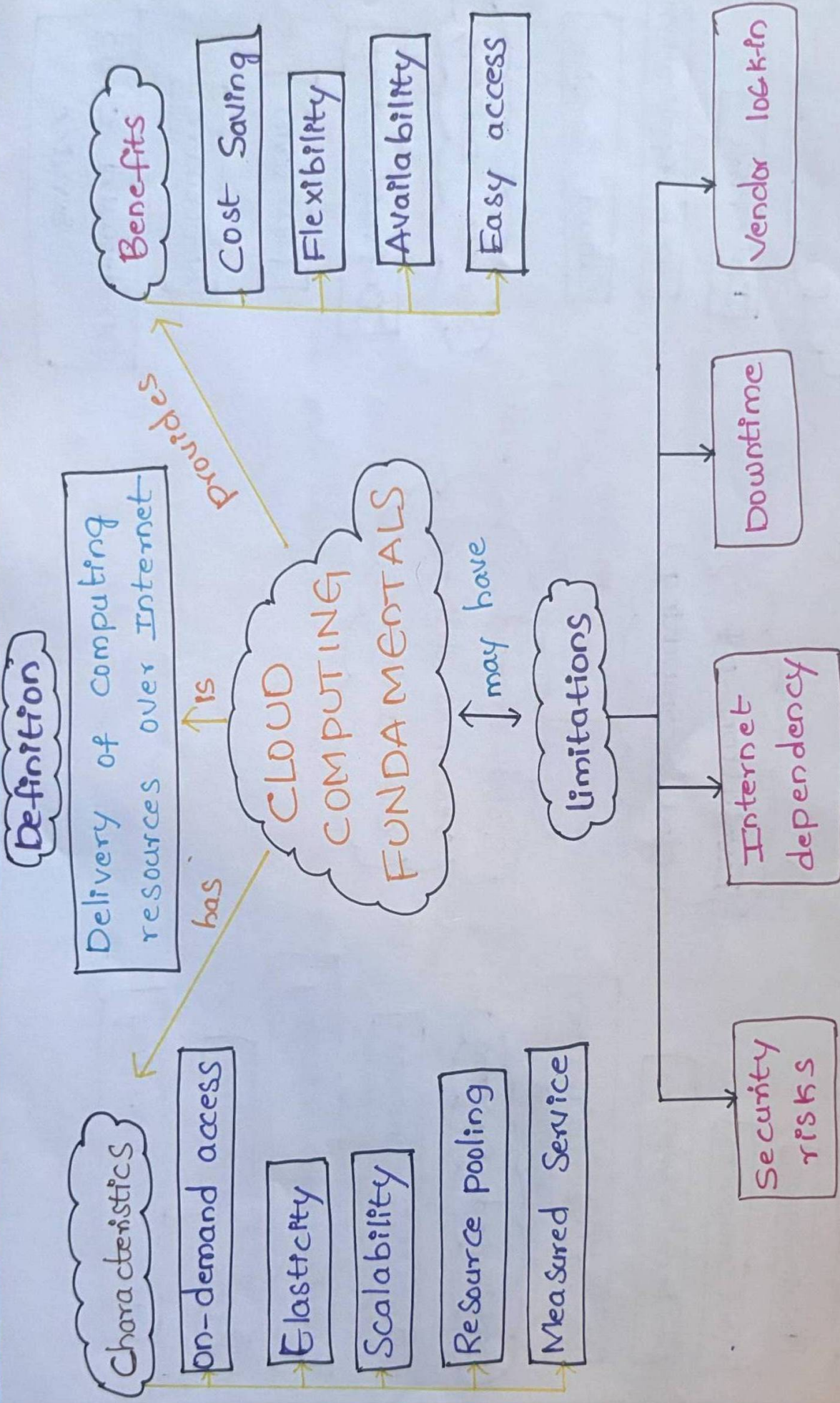
- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer □ Infrastructure layer, platform layer, application layer, shared responsibility.

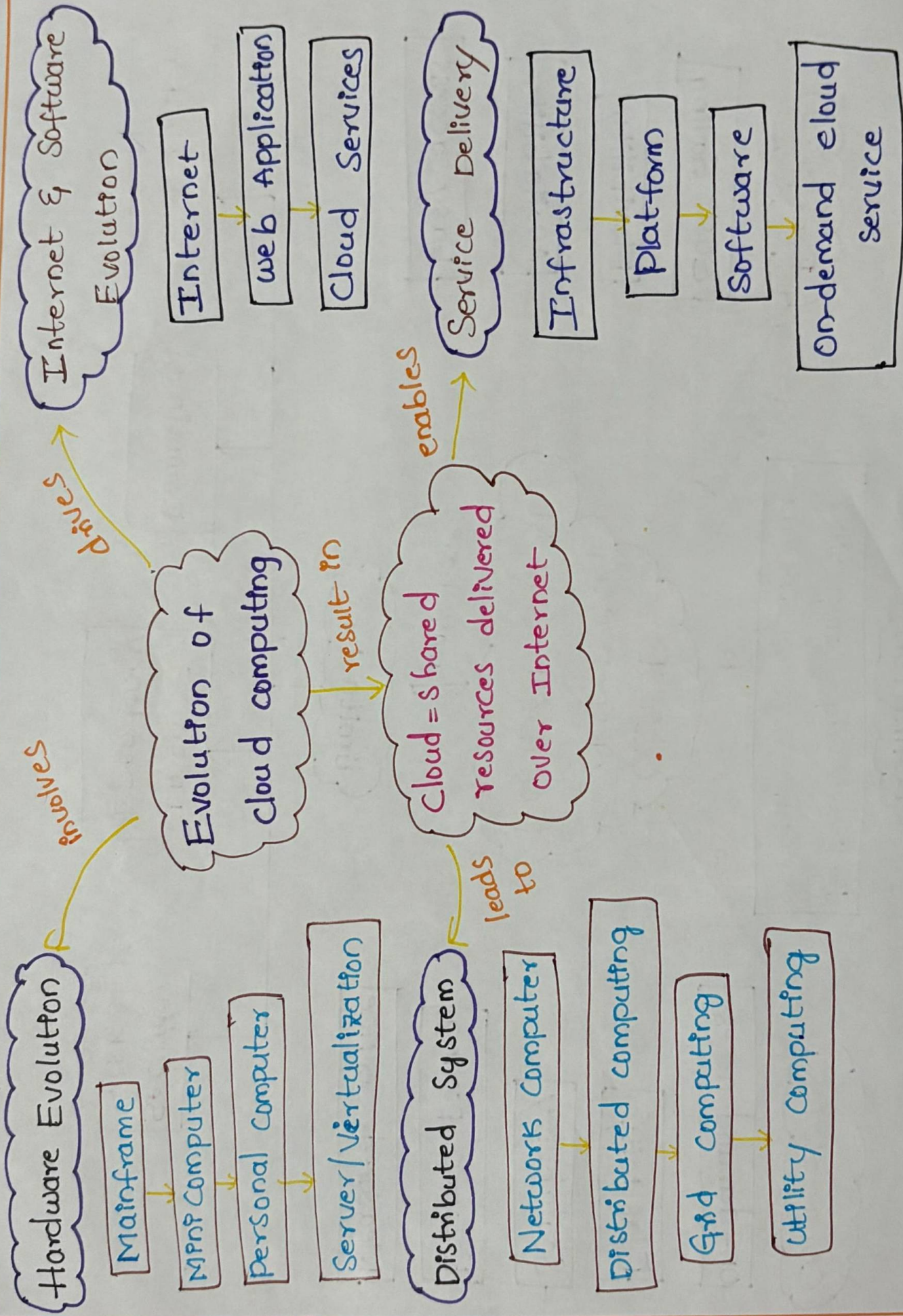
Concept Mapping Rubric:

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

1)





Service Models of Cloud Computing

IaaS - Infrastructure as a Service

Provides

- Provider manages
- Hardware
 - Software
 - Networking
 - Virtualization

User manages

- User Manages
- OS
 - Applications
 - Data

PaaS - Platform as a Service

Provides

- Provider manages
- Infrastructure
 - Runtime
 - Middleware

User manages

- User Manages
- Application Code
 - Data

SaaS - Software as a Service

Provides

- Provider manages
- Application
 - Platform
 - Infrastructure
 - Maintenance

User manages

- User manages
- Data
 - Configuration
 - Usage

XaaS - Anything as a Service

Provides

- Provider Manages
- Anything as a Service

User manages

- User manages
- Storage
 - Database
 - Security
 - AI

Virtualization and Web Services as enablers of cloud

